

CS & IT ENGINEERING



Data Structure & Programming

Stack and Queues

Lecture No.- 02



By- Satya sir

Recap of Previous Lecture



- Time Complexity cheat sheet of LL

- Stack DS

- Nature
- Push, Pop, Peek
- Examples



Topics to be Covered



- H/w & Solving
- Stack Permutations
- Applications of stack
 - | Expression Conversion



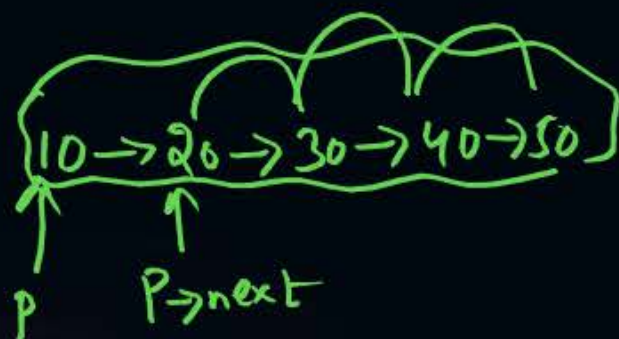


Topic : Linked Lists



Consider the function f defined below.

```
struct item
{
    int data;
    struct item * next;
};
int f(struct item *p)
{
    return ((p == NULL) || (p->next == NULL) ||
            ((p->data <= p->next->data) &&
             f(p->next)));
}
```



$P == \text{NULL} \Rightarrow \text{Exp1}$

$P \rightarrow \text{next} == \text{NULL} \Rightarrow \text{Exp2}$

$P \rightarrow \text{data} \leq P \rightarrow \text{next} \rightarrow \text{data} \Rightarrow \text{Exp3}$

$f(P \rightarrow \text{next}) \Rightarrow \text{Exp4}$

Elements must be in increasing order

$\text{Exp1} \ || \ \text{Exp2} \ || \ (\text{Exp3} \ \&\& \ \text{Exp4})$

$\text{True} \ \&\& \ \text{True} \Rightarrow \text{True}$

For a given linked list p, the function f returns 1 if and only if

- (A) the list is empty or has exactly one element
- ☒ (B) the elements in the list are sorted in non-decreasing order of data value
- (C) the elements in the list are sorted in non-increasing order of data value
- (D) not all elements in the list have the same data value.

GATE 2003



Topic : Stack Operations, Permutations



Stack Permutations

When 'n' elements Pushed in Dynamic sequence, Then Total No. of op Permutations will be $n!$

- Insertion/Push into Stack can be done in either Static/fixed order [or] Dynamic/variable order.
- Then The Total Possible Permutations of Output is Stack Permutations.

Ex: Consider 3 elements A, B, C Pushed into Stack in any order.
Then Total how many Possible op sequences?

①
Push A
Push B
Push C
Pop
Pop
Pop
op: C, B, A

②
Push A
Pop
Push B
Pop
Push C
Pop
A, B, C

③
Push A
Pop
Push C
Pop
Push B
Pop
A, C, B

④
Push B
Push A
Push C
Pop
Pop
Pop
C, A, B

Push B
Push C
Push A
Pop
Pop
Pop
A, C, B

⑤
Push B
Push C
Pop
Pop
Push A
Pop
C, B, A

⑥
Push B
Pop
Push C
Pop
Push A
Pop
B, C, A

⑦
Push B
Pop
Push A
Pop
Push C
Pop
B, A, C

$= 3!$
6 sequences

A, B, C
A, C, B
B, C, A
B, A, C
C, B, A
C, A, B



Topic : Stack Operations, Permutations

$${}^nC_r = \frac{n!}{(n-r)! r!}$$



Push in static/fixed sequence

- When 'n' Elements Pushed into Stack in some fixed or static Sequence, Then Total number of

o/p Permutations will be $\frac{{}^{2n}C_n}{n+1}$ nth Catalan number.

$$\frac{{}^{2n}C_n}{n+1} \Rightarrow \frac{{}^6C_3}{4} \Rightarrow \frac{6!}{3! \times 3!} * \frac{1}{4} = \frac{6 \times 5 \times 4 \times 3!}{3! \times 3!} * \frac{1}{4} = \underline{\underline{5}}$$

Ex: Let 3 Elements A, B, C Pushed in the same sequence.
Then how many o/p Permutations Possible?

1) Push A Pop Push B Pop Push C Pop A, B, C	2) Push A Push B Push C Pop Pop Pop C, B, A	3) Push A Push B Pop Push C Pop Pop B, C, A	4) Push A Push B Pop Pop Push C Pop B, A, C	5) Push A Pop Push B Push C Pop Pop A, C, B
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C, A, B Sequence is Not Possible/Valid

Push A
Push B
Push C
Pop — C
Pop — B
Pop — A

Push B
Push A
Push C
Pop
Pop
Pop
C, A, B

Push C
Pop
Push A
Pop
Push B
Pop
C, A, B

Push B
Push C
Pop
Push A
Pop
Pop
C, A, B

Given order of insertion is not followed.



Topic : Stack Operations, Permutations



#Q. Consider an Empty Stack S . If Elements $1, 2, 3, 4, 5$ are inserted in the same order, Then which of the below combination is Invalid?

A) ✓ $4, 5, 3, 2, 1$

B) ✓ $3, 2, 4, 5, 1$

C) ✗ $1, 3, 5, 2, 4$

D) ✓ $2, 3, 1, 4, 5$

Push $1, 2, 3, 4$, Pop, Push 5 , Pop, Pop, Pop, Pop

Push $1, 2, 3$, Pop, Pop, Push 4 , Pop, Push 5 , Pop, Pop

Push 1 , Pop, Push $2, 3$, Pop, Push $4, 5$, Pop, Pop, Pop $[1, 3, 5, 4, 2]$

Push $1, 2$, Pop, Push 3 , Pop, Pop, Push 4 , Pop, Push 5 , Pop



Topic : Stack Operations, Permutations



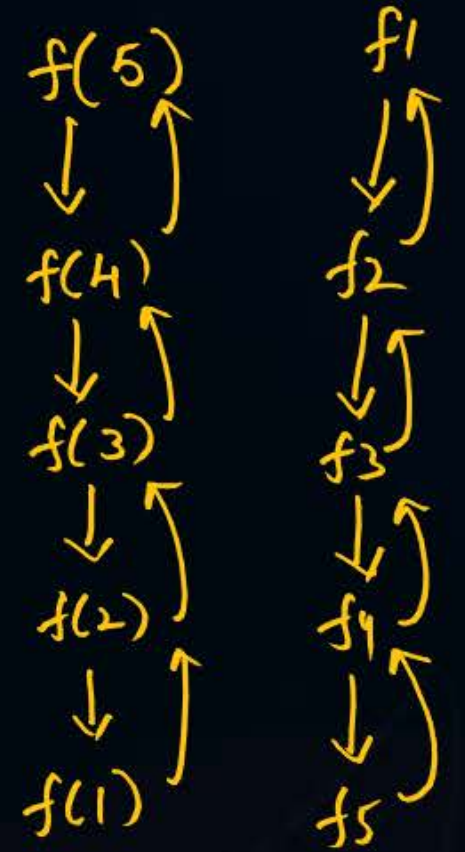
#Q. Consider an Empty stack S. Identify Incorrect Pair of (i/p, o/p) Sequence.

- | | <u>i/p</u> | <u>o/p</u> | |
|-----|--------------------|----------------------------|---|
| A ✓ | (6, 4, 3, 2, 5, 1) | (2, 3, 5, 4, 1, 6) | Push 6, 4, 3, 2, ² POP, ³ POP, Push 5, ⁵ POP, ⁴ POP, Push 1, ¹ POP, ⁶ POP |
| B ✓ | (3, 2, 1, 4, 5, 6) | (1, 5, 4, 6, 2, 3) | Push 3, 2, 1, ¹ POP, Push 4, 5, ⁵ POP, ⁴ POP, Push 6, ⁶ POP, ² POP, ³ POP |
| C ✗ | (4, 3, 6, 1, 2, 5) | (6, 5, 2, 1, <u>4</u> , 3) | Push 4, 3, 6, ⁶ POP, Push 1, 2, 5, ⁵ POP, ² POP, ¹ POP, ³ POP, ⁴ POP |
| D ✓ | (5, 1, 3, 2, 4, 6) | (2, 4, 3, 6, 1, 5) | Push <u>5</u> , 1, 3, 2, ² POP, push <u>4</u> , ⁴ POP, ³ POP, Push <u>6</u> , ⁶ POP, ¹ POP, ⁵ POP |



Applications of Stack

- ① Undo operation
- ② Recursion Implementation
- ③ Function Implementation
- ④ Non-Maskable Interrupt Service
- ⑤ Tree/Graph Traversals - Depth First Traversal
- ⑥ Browsing history
- ⑦ Expression Conversion
- ⑧ Expression Evaluation
- ⑨ Backtracking
- ⑩ Bracket Balancing
- ⑪ Redo operation
- ⋮





Topic : Expression Conversion



Expression : Any executable statement in an algorithm / Program

Types of Expressions

- 1) Input Expression : `scanf()`, `cin <<`, `Scanner class`, `input()`...
- 2) Output Expression : `printf()`, `putc()`, `cout`, `Print()`
- 3) Declaration statement : `int a, b; float c; char s[]`...
- 4) Initialization Statement : `int a=5; float b=1.317`...
- 5) Conditional Expression : `if`, `if-else`, `switch`, `match-case`
- 6) Iterative Expression : `do-while`, `while`, `for`
- 7) Unconditional Expression : `break`, `continue`, `return`, `Pass`...
- 8) Function Definitions, calling : `void f() { }`, `f()`;
- 9) Computational statements : `a = b + c * d / e` ✓

In computational stmt, any operator can be used.

But, we consider Arithmetic operators

Default Priority $[+, -, *, /, \wedge]$ Exponentiation

① $\wedge$	} Associativity Left to Right
② $*, /$	
③ $+, -$	



Arithmetic Expression

Default Precedence : $\wedge > (*, /) > (+, -)$

Ex: $a = 5 + 4 * 3 - 2 \rightarrow \text{No. of tokens} = 9$

As per default Precedence :
 $4 * 3 = 12$
 $5 + 12 = 17$
 $17 - 2 = 15$

But, if it needs to be evaluated as $\left. \begin{array}{l} 5 + 4 = 9 \\ 3 - 2 = 1 \\ 9 * 1 = 9 \end{array} \right\} \Rightarrow \text{Expression should be written as } \underline{a = (5 + 4) * (3 - 2)}$
No. of tokens = 13



Topic : Expression Conversion



An Expression can be written in 3 forms :

① Infix Notation $\Rightarrow$ operand1 operator operand2

② Prefix " $\Rightarrow$ operator operand1 operand2

③ Postfix " $\Rightarrow$ operand1 operand2 operator

Example

$a+b$

$+ab$

$ab+$

Expression conversion ? $\longrightarrow$ To be discussed ... on Monday



2 mins Summary



- Stack Permutations
- Applications of Stack
 - Exp Conversion

To be Contd....





THANK - YOU